

FPGA Prototyping of converters, and Simulation of Totem-Pole PFC–LLC Converter For EV Charging

Master's Thesis Project Phase I

Report

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Abstract

This research presents a comprehensive investigation into the design, modeling, and real-time validation of advanced power converter topologies for high-efficiency electric vehicle charging systems. The work encompasses three interconnected studies: (1) a market-driven survey of contemporary electric vehicle architectures and competitive landscape analysis, identifying key trends in fast-charging infrastructure and converter technology adoption; (2) systematic simulation and comparative analysis of industry-standard Totem-Pole Power Factor Correction (PFC) and LLC resonant converter topologies in MATLAB-Simulink, evaluating performance metrics including efficiency, transient response, and harmonic distortion across representative operating conditions; and (3) FPGA-based hardware-in-the-loop (HIL) real-time validation of discrete-time digital controllers, demonstrating seamless integration between simulation environments and embedded implementations. The converter platform operates at 7.7 kW output power with 400 V ports, incorporating soft-switching characteristics and galvanic isolation to achieve reduced electromagnetic interference. A Z-domain compensator is derived through systematic discretization of an S-domain reference controller, enabling precision voltage regulation with fast transient settling and robust operation across dynamic load variations. Experimental HIL results confirm superior damping and minimal overshoot characteristics of the discrete-time implementation relative to continuous-time equivalents.

Keywords: Totem-Pole PFC, LLC resonant converter, FPGA-based digital control, on-board charger, soft switching, SiC/GaN, EV charging, market survey, MATLAB simulation.

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1 Introduction

The need for effective and small power conversion systems has increased due to the world's shift to electric vehicles. As a vital link between the grid and the car battery, electric vehicle (EV) chargers must have strong control, low harmonic distortion, and high efficiency. Modern charger topologies sometimes use multi-stage designs to satisfy these criteria, with a DC–DC stage providing controlled battery charging and a Power Factor Correction (PFC) stage guaranteeing grid standard compliance. Because of their exceptional efficiency and applicability for a broad power range, the Totem-Pole PFC and LLC Resonant Converter have become popular options. Despite their speed and maturity, traditional analog controllers have limits with regard to digital communication, adaptability, and flexibility. Field-Programmable Gate Arrays (FPGAs), on the other hand, offer a reconfigurable digital control platform that can integrate protection and monitoring circuitry into a single chip, perform high-speed parallel operations, and execute real-time control algorithms. For sophisticated power converters, this makes FPGA-based control a desirable choice.

With an initial focus on a buck converter prototype as a learning and validation platform, this project seeks to provide a unified FPGA-oriented digital control framework for power converters. The goal is to comprehend the difficulties in real-time implementation, the discretization of continuous-time controllers, and the synchronization of control loops using digital pulse-width modulation (DPWM). The knowledge gathered from this initial phase will be used to intricate converter systems like the Full-Bridge LLC resonant converter and Totem-Pole PFC.

In order to comprehend current developments in EV charger design and digital control technologies, the study also involves a market-oriented survey. The project intends to close the gap between theoretical control design and real-world hardware implementation by fusing simulation, theoretical analysis, and FPGA-based prototyping.

2 Literature Review

2.1 Global Trends in EV Charger Design

Recent advancements in electric vehicle (EV) charger design reveal a clear shift toward high-efficiency, compact, and modular architectures. Modern fast and ultra-fast chargers typically employ a two-stage configuration, where the first stage (AC–DC) performs active rectification with power factor correction (PFC) to comply with IEC 61000-3-2 harmonic standards, and the second stage (DC–DC) provides tightly regulated high-voltage DC output [5]. Studies such as those by Suárez et al. [1] and the Texas Instruments Application Note SLLA497 [5] highlight that efficiencies exceeding 96% are routinely achieved in modular converter systems rated from 7 kW to over 350 kW. Among various PFC architectures, the Totem-Pole PFC converter has emerged as the preferred single-phase topology for onboard chargers up to 11 kW, particularly when realized using wide bandgap devices such as silicon carbide (SiC) and gallium nitride (GaN) [3, 8]. This topology delivers efficiency greater than 99% and supports high switching frequencies in the 200–400 kHz range, outperforming traditional boost-based PFC designs in both performance and power density [3]. For the isolated DC–DC conversion stage, the LLC resonant converter remains the dominant choice due to its soft-switching capability, high efficiency, and excellent electromagnetic interference (EMI) characteristics [4, 9]. By employing frequency modulation to achieve zero-voltage switching (ZVS) on the primary side and zero-current switching (ZCS) on the secondary side, LLC converters routinely achieve efficiencies above 97% for power levels in the range of 10–20 kW [4, 9, 11]. The combination of a Totem-Pole PFC front end and an LLC resonant DC–DC stage therefore represents the state-of-the-art solution for high-performance EV charger systems, balancing efficiency, reliability, and compactness [8, 9].

2.2 Market Survey of Electric Vehicle Charging Hardware

This section gives a concise survey of leading electric vehicles (EVs) in four main groups:

1. Indian Passenger Cars
2. Indian Commercial / 3-Wheel / LCV EVs
3. Global Passenger EVs
4. Global Commercial EVs

For each group, summarized tables show the top 10 models and compare their onboard charger ratings, DC fast-charging power, charging standards, battery voltages, and typical PFC and converter topologies used [7].

2.2.1 Top 10 Indian Passenger EVs

Table 2.1: Top 10 Indian Passenger Electric Vehicles and Their Charging Characteristics [7]

No.	Model	OBC (kW)	DC Fast (kW)	Charging Standard	Nominal Voltage (V)	Typical Converter / PFC Topology
1	Tata Nexon.ev	7.2	50–100	CCS2 / Type-2	350–400	Interleaved boost PFC + LLC/PSFB DC–DC
2	Tata Tiago.ev	3.3 / 7.2	25	CCS2 / Type-2	300–400	Single-phase boost PFC + LLC DC–DC
3	Tata Punch.ev	7.2	50	CCS2 / Type-2	350–400	Two-stage: Boost PFC + LLC DC–DC
4	MG ZS EV	7.4	50	CCS2 / Type-2	400	Boost PFC + LLC/PSFB
5	BYD Atto 3	7.2	50–150	CCS2 / Type-2	400	Active PFC + Isolated LLC DC–DC
6	Mahindra XUV.ev	7.2–11	50–100	CCS2 / Type-2	400	Boost PFC + PSFB DC–DC
7	Hyundai Creta EV / Kona	7.2–11	50–100	CCS2 / Type-2	400	Two-stage OBC (PFC + LLC DC–DC)
8	Kia EV6 (India import)	11	240–350	CCS2 / Type-2	800	Three-phase active rectifier + DAB DC–DC
9	MG Comet EV	3.3	—	Type-2 AC	300	Simple AC–DC single-stage charger
10	Mahindra e-Verito	3.3–7.2	25–50	CCS2 / Type-2	300–400	Boost PFC + LLC DC–DC

2.2.2 Top 10 Indian Commercial / LCV / 3-Wheeler EVs

Table 2.2: Top 10 Indian Commercial and 3-Wheeler Electric Vehicles

No.	Model	OBC (kW)	DC Fast (kW)	Standard	Nominal Voltage (V)	Typical Converter / PFC Topology
1	Tata Ace EV	2–3	25–30	Proprietary DC / AC plug	93	Low-voltage single-stage charger, DC fast off-board PSFB
2	Mahindra Treo Zor	2–3	—	AC plug	48	Simple AC–DC rectifier, no PFC
3	Piaggio Ape E-City	2–3	—	AC plug	51	Single-stage low-voltage rectifier
4	Bajaj RE E-TEC 9.0	3	—	AC plug	72	Simple onboard charger (rectifier + DC–DC)
5	Olectra Mini E-Bus	7–11	50–150	CCS2	400	Active PFC + PSFB DC–DC
6	Tata Ultra EV / Yodha EV	7–11	50–150	CCS2	400	Three-phase PFC + LLC DC–DC
7	Switch Mobility LCVs	11–22	100–150	CCS2	400	Three-phase active rectifier + DAB DC–DC
8	Mahindra e-KUV Fleet	3.3–7.2	25–50	CCS2	300–400	Boost PFC + LLC DC–DC
9	JBM / Ashok Leyland Small Bus	11–22	100–150	CCS2	400	Three-phase PFC + DAB DC–DC
10	Kinetic Safari EV / TVS Cargo	2–3	—	AC plug	48–72	Single-stage AC–DC with limited PFC

2.2.3 Top 10 Global Passenger EVs

Table 2.3: Top 10 Global Passenger EVs and Their Charger Configurations

No.	Model	OBC (kW)	DC Fast (kW)	Standard	Pack Voltage (V)	Typical Converter / PFC Topology
1	Tesla Model Y	11	250–300	NACS / CCS2	350–400	Active rectifier + LLC/PSFB DC–DC
2	Tesla Model 3	11	250–300	NACS / CCS2	350–400	Active PFC + LLC DC–DC
3	BYD Dolphin	7	65	GB/T / CCS2	350–400	Boost PFC + LLC DC–DC
4	BYD Seal	11	150	GB/T / CCS2	400	Active rectifier + LLC/PSFB
5	BYD Atto 3	7–11	80	GB/T / CCS2	400	Two-stage OBC (PFC + LLC)
6	Wuling MINI EV	2–6	—	GB/T	200–350	Single-stage rectifier
7	VW ID.4	11	125	CCS2	400	Boost / Active PFC + LLC DC–DC
8	Hyundai Ioniq 5	11	350	CCS2	800	Totem-pole PFC + DAB/CLLC DC–DC
9	Kia EV6	11	240–350	CCS2	800	Active rectifier + DAB DC–DC
10	MG4 EV	7–11	100	CCS2	400	Two-stage OBC (PFC + LLC)

2.2.4 Top 10 Global Commercial / LCV EVs

Table 2.4: Top 10 Global Commercial Electric Vans and Their Charger Configurations

No.	Model	OBC (kW)	DC Fast (kW)	Standard	Nominal Voltage (V)	Typical Converter / PFC Topology
1	Ford E-Transit	11.3	115	CCS2	400	3-Phase active rectifier + PSFB DC–DC
2	Mercedes eSprinter	11–22	115	CCS2	400	3-Phase PFC + LLC DC–DC
3	Renault Kangoo E-Tech	11–22	80	CCS2	400	Single/3-Phase PFC + LLC
4	VW e-Crafter	11	80	CCS2	400	Boost PFC + PSFB
5	BYD eT3 Van	7.2	60–80	GB/T / CCS2	400	Two-stage OBC (Boost + LLC)
6	Maxus eDeliver 3	7–11	80	CCS2	400	Interleaved PFC + LLC
7	Nissan Townstar / eNV200	11–22	50	CCS2	400	Two-stage OBC (PFC + LLC)
8	Stellantis Small Vans	11–22	50–100	CCS2	400	3-Phase Active Rectifier + PSFB
9	BrightDrop Zevo 600	11.5 / 19.2	120	CCS1 / 2	400	Active PFC + LLC DC–DC
10	Rivian EDV Fleet Van	11	100	CCS1 / 2	400	Active rectifier + LLC/DC–DC

2.3 Most Common Converter Topologies Used in EV Chargers

When looking at a wide range of electric vehicles (EVs) in India and around the world, some clear patterns stand out in how onboard chargers are built:

Two-stage design is the norm: Almost every EV uses a charger made of two parts. First is a power factor correction (PFC) section that makes sure the charger draws electricity smoothly from the wall, minimizing “bad” current that could mess with the grid. Second is a DC-DC converter that takes this electricity and turns it into precisely-controlled high-voltage power for the battery. This design keeps things efficient, safe, and compliant with regulations [7, 5].

How PFC is done: For typical cars that use home plugs (3–7 kW chargers), manufacturers often use something called an interleaved boost PFC—basically, a smart way of smoothing power using several circuits working together [3]. For bigger chargers (11–22 kW, often three-phase), automakers often use advanced systems like totem-pole PFC or three-phase active rectifiers [8]. These can handle higher loads and work better with new semiconductor materials.

The “engine room” for battery charging: After PFC, it’s up to the DC-DC converter to do the heavy lifting. Most modern EVs rely on LLC resonant converters for 3–11 kW systems because they are highly efficient and create less electrical noise [4, 9]. For faster, more powerful chargers (11–22 kW and up, sometimes for large vehicles or super-fast charging), designers like phase-shifted full bridge (PSFB) and dual active bridge (DAB) topologies—they can handle more current and sometimes run power both ways (sending energy from vehicle to grid) [12, 11].

Small EVs, simpler chargers: Rickshaws, three-wheelers, and light commercial vehicles with smaller batteries (typically under 100 V) usually use simpler chargers: a basic one-step circuit, sometimes without any PFC, because their power needs are low [7].

What global brands do: Big companies like Tesla, BYD, VW, Hyundai, Ford, and Mercedes all use two-stage chargers. The main differences between their chargers depend on how much power is needed and what kind of semiconductor (silicon or newer SiC) they use—this affects speed, cost, and energy savings [1, 7].

3 FPGA-Oriented Design and Realization of Power Converters

This section focuses on understanding and implementing power converter control through FPGA-based digital prototyping. The aim is to develop a unified digital control framework capable of handling both the Totem-Pole PFC and Full-Bridge LLC resonant converter stages. Initially, a buck converter prototype is implemented to understand controller discretization, timing synchronization, and hardware testing. The same control principles are later extended to higher-level topologies for improved efficiency and real-time validation.

3.1 FPGA-Based Digital Control and Prototyping

3.1.1 Motivation for FPGA in Converter Control

Traditional analog controllers—implemented using op-amps and RC networks—offer simplicity but lack flexibility, precision, and adaptability. As converter topologies become more complex and operate at higher switching frequencies, the control system must address:

- rapid transient response and parameter variations,
- multi-loop voltage and current regulation,
- real-time modulation and fault protection, and
- integration with diagnostics and communication interfaces.

While microcontrollers (MCUs) and digital signal processors (DSPs) can execute control algorithms, their sequential instruction flow introduces latency and limits achievable bandwidth. In contrast, **FPGAs** provide:

1. **Deterministic Timing:** Dedicated hardware logic enables sub-microsecond loop updates.
2. **True Parallelism:** Current and voltage loops execute concurrently.
3. **High-Resolution PWM:** Duty cycles adjustable with 12–16-bit precision, achieving switching resolution below 100 ns.
4. **Rapid Prototyping:** MATLAB/Simulink or HDL-based controllers can be synthesized for real-time execution.
5. **Hardware-in-the-Loop (HIL) Testing:** Converter dynamics can be emulated to validate control logic.
6. **Reconfigurability:** New compensators or algorithms can be added via bitstream updates.

Hence, the FPGA bridges the gap between simulation and hardware realization, acting as both a *research platform* and a *production-grade controller*.

3.1.2 Discretization of Analog Controllers

Digital realization requires conversion of the continuous controller $G_c(s)$ into its discrete form $G_c(z)$. The Tustin (bilinear) transformation provides an effective way to maintain phase and gain accuracy near the desired crossover frequency:

$$s = \frac{2}{T_s} \frac{1 - z^{-1}}{1 + z^{-1}}.$$

This transformation preserves frequency-domain characteristics, making it suitable for precise converter control.

Table 3.1: Common discretization methods

Method	Substitution	Remarks
Forward Euler	$s = \frac{z-1}{T_s}$	Simple, introduces phase lag
Backward Euler	$s = \frac{1-z^{-1}}{T_s}$	Stable but attenuates HF
Tustin (Bilinear)	$s = \frac{2}{T_s} \frac{1-z^{-1}}{1+z^{-1}}$	Best frequency accuracy
Matched Z-Transform	Exact pole-zero mapping	Analytically complex

The Tustin method is adopted for both Type-II and Type-III compensator design in this study.

3.1.3 Control Exercise: Discretization and Implementation

Objective To discretize and implement a continuous-time controller on FPGA using MATLAB/Simulink HDL Coder or hand-coded HDL, and to compare the analog and digital frequency responses.

Procedure

1. Define $G_c(s) = K_c \frac{(1+s/\omega_z)}{s(1+s/\omega_p)}$.
2. Choose $T_s = 1/f_{PWM}$ (e.g., 20 μ s for 50 kHz).
3. Apply the Tustin substitution.
4. Compute digital coefficients using MATLAB's `c2d()`.
5. Implement in Simulink using the Discrete Transfer Function block.
6. Verify Bode alignment up to $0.1 \times f_s$.
7. Generate HDL and deploy to FPGA.
8. Interface with DPWM and record output voltage/current response.

3.1.4 Converter Parameters

- Input Voltage (V_{in}): 30 V
- Inductor (L): 200 μH
- Output Capacitor (C): 400 μF
- ESR of Capacitor (r): 0.1 Ω
- Load Resistance (R): 5 Ω
- Switching Frequency (f_{sw}): 100 kHz
- Desired Crossover Frequency (f_c): 5 kHz for voltage-mode (Type-III), 1 kHz for current-mode (Type-II)
- Desired Phase Margin: 90 to 120°

3.1.5 Circuit Diagram

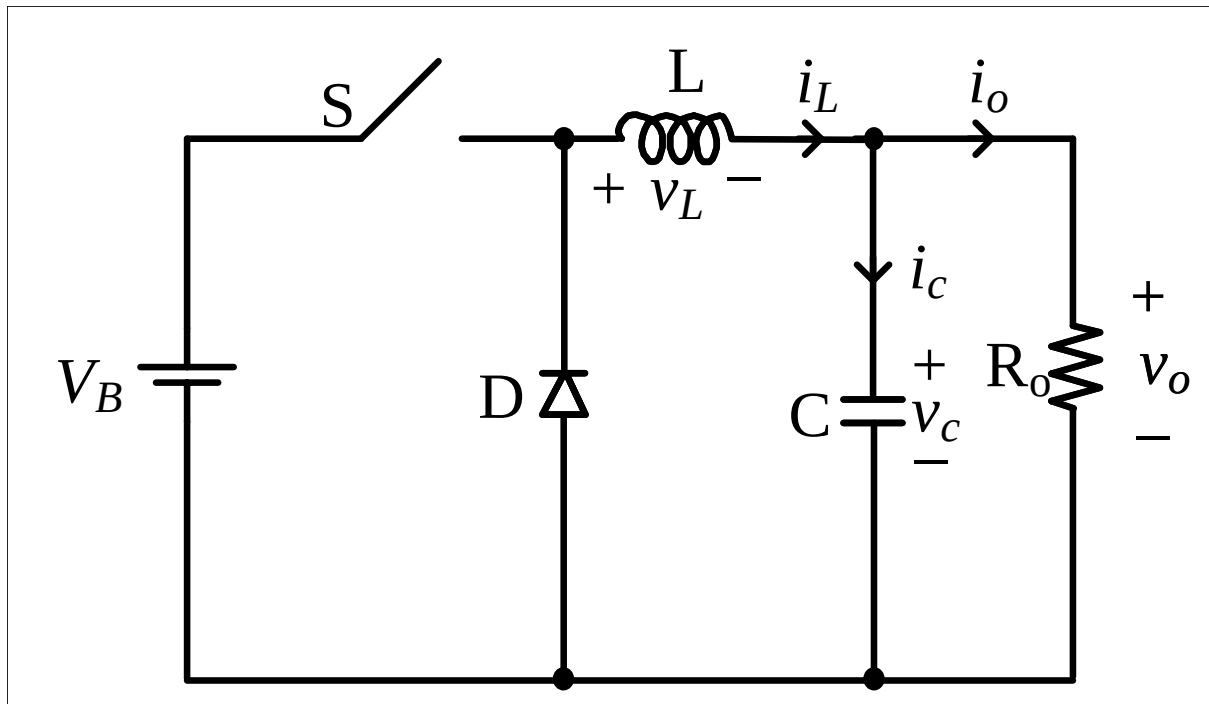


Figure 3.1: Buck Converter Circuit Diagram with Compensator

Accurate control of a buck converter requires dynamic modeling, enabling derivation of the small-signal transfer function which relates the output voltage to the control variables. The output voltage is influenced by the input voltage V_{in} and the duty cycle d . The transfer function under small-signal conditions forms the basis for compensator design.

3.1.6 Voltage-Mode Control (Type-III Compensator)

The control-to-output transfer function for the buck converter is:

$$G_p(s) = \frac{V_{in}}{LC} \cdot \frac{1 + sCr}{s^2 + s\left(\frac{r}{L} + \frac{1}{RC}\right) + \frac{1}{LC}} \quad (1)$$

Type-III Compensator Design (k-factor method)

1. **Bode Plot:** Use MATLAB to simulate at $f_c = 5$ kHz.

2. **Unity gain:**

$$G_{MB} = \frac{1}{|G_p|_{f_c}} \quad (2)$$

3. **Phase boost:**

$$\text{Phase boost} = PM - \angle G_p|_{f_c} - 90^\circ \quad (3)$$

4. **k-factor:**

$$k = \tan\left(45^\circ + \frac{\text{Phase boost}}{4}\right) \quad (4)$$

5. **Zero and Pole locations:**

$$f_z = \frac{f_c}{k} \quad f_p = f_c \cdot k \quad (5)$$

6. **Type-III Compensator:**

$$G_c(s) = \frac{K_c}{s} \cdot \frac{\left(1 + \frac{s}{\omega_{z1}}\right)^2}{\left(1 + \frac{s}{\omega_{p1}}\right)^2} \quad (6)$$

where $\omega_{z1} = 2\pi f_z$, $\omega_{p1} = 2\pi f_p$, and K_c sets unity gain at crossover.

3.1.7 Current-Mode Control (Type-II Compensator)

The simplified plant transfer function for current-mode is:

$$G_p(s) = R \cdot \frac{1 + srC}{1 + sRC} \quad (7)$$

Type-II Compensator Design

1. Desired crossover $f_c = 1$ kHz, phase margin $PM = 60^\circ$.

2. **Phase boost:**

$$\text{Phase boost} = PM - \angle G_p|_{f_c} - 90^\circ \quad (8)$$

3. **k-factor:**

$$K_{boost} = \tan\left(45^\circ + \frac{\text{Phase boost}}{2}\right) \quad (9)$$

4. Zero and Pole frequencies:

$$f_z = \frac{f_c}{K_{boost}} \quad f_p = f_c \cdot K_{boost} \quad (10)$$

5. Type-II Compensator:

$$G_c(s) = \frac{k_c}{s} \frac{1 + \frac{s}{\omega_z}}{1 + \frac{s}{\omega_p}} \quad (11)$$

where $\omega_z = 2\pi f_z$, $\omega_p = 2\pi f_p$.

MATLAB Implementation The MATLAB scripts were developed to perform both the analog and digital compensator design, discretization, and validation. Separate scripts are used for Type-III (voltage-mode) and Type-II (current-mode) control.

MATLAB Results and Design Parameters Key results and parameters obtained from MATLAB are summarized below.

Voltage-Mode (Type-III) Compensator Results

Zero frequency (fz): 1902.68 Hz
Pole frequency (fp): 13139.33 Hz
Compensator gain (kc): 7364.8612
k-boost factor: 2.6279
Phase boost: 96.67 degrees

Current-Mode (Type-II) Compensator Results

Zero frequency (fz): 452.12 Hz
Pole frequency (fp): 2211.79 Hz
Compensator gain (kc): 6946.1946
Kboost factor: 2.2118
Phase boost: 41.34 degrees

Bode Plots and Transfer Function Analysis Bode plots compare the analog compensator $G_c(s)$, plant $G_p(s)$, open-loop $G_p(s)G_c(s)$, and discretized digital compensator $G_c(z)$. The close correspondence between analog and digital phase margins validates the accuracy of Tustin-based discretization.

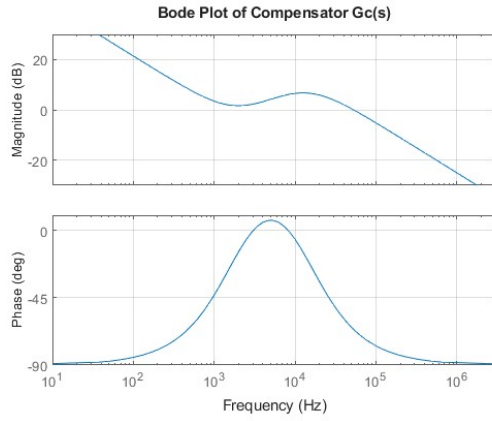


Figure 3.2: Bode plot of Compensator $G_c(s)$ (Voltage-Mode Control).

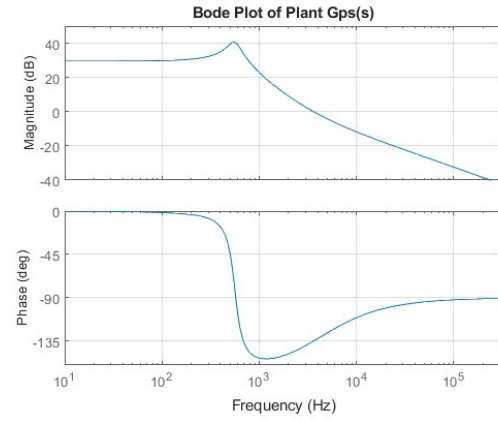


Figure 3.3: Bode plot of Plant $G_p(s)$ (Voltage-Mode Control).

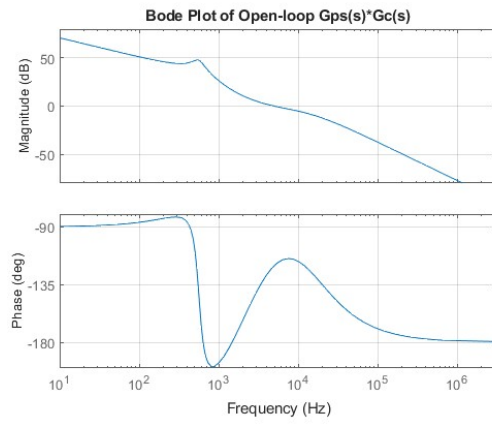


Figure 3.4: Bode plot of Open-loop $G_p(s) \times G_c(s)$ (Voltage-Mode Control).

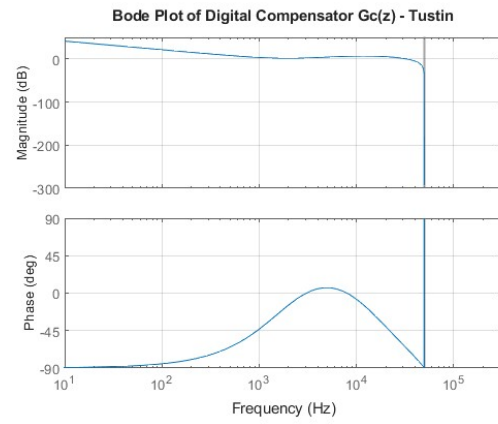


Figure 3.5: Bode plot of Digital Compensator $G_c(z)$ (Voltage-Mode Control).

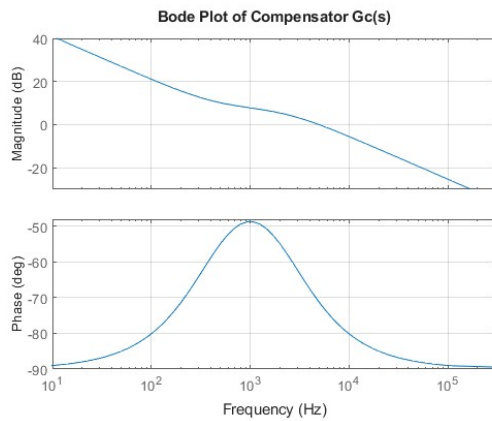


Figure 3.6: Bode plot of Compensator $G_c(s)$ (Current-Mode Control).

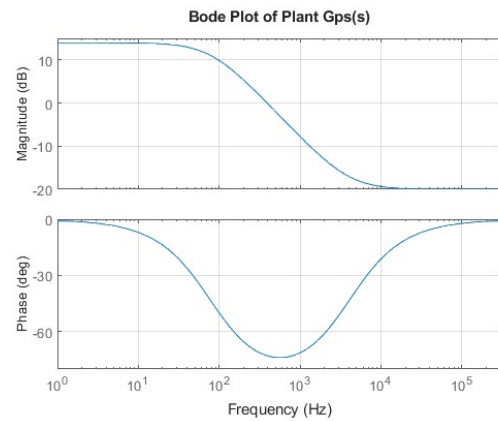


Figure 3.7: Bode plot of Plant $G_p(s)$ (Current-Mode Control).

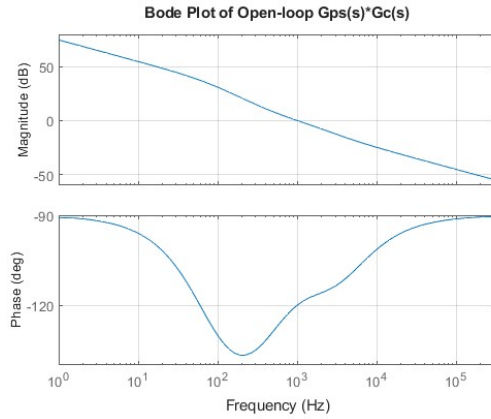


Figure 3.8: Bode plot of Open-loop $G_p(s) \times G_c(s)$ (Current-Mode Control).

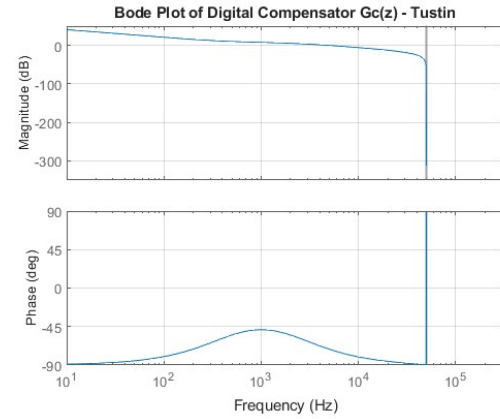


Figure 3.9: Bode plot of Digital Compensator $G_c(z)$ (Current-Mode Control).

Summary of Observations Both Type-II and Type-III digital compensators achieved the desired phase margin and bandwidth. The Tustin method maintained phase accuracy up to $0.1 f_s$, confirming minimal distortion after discretization. Voltage-mode control demonstrated faster transient recovery and negligible steady-state error, while current-mode control exhibited improved load disturbance rejection. The results confirm the feasibility of deploying MATLAB-designed digital compensators on FPGA hardware without significant loss in dynamic performance.

3.1.8 FPGA Implementation Flow

The FPGA implementation follows these sequential stages:

1. **Modeling in Simulink:** Develop the plant $G_p(s)$ and compensator $G_c(s)$.
2. **Discretization:** Convert $G_c(s)$ to $G_c(z)$ using the Tustin transformation.
3. **PWM Generation:** Compute duty ratio $D[n] = y[n]/V_{ref}$ to drive the DPWM comparator.
4. **HDL Generation:** Export the discrete model as synthesizable VHDL/Verilog using HDL Coder.
5. **HIL Testing:** Run the FPGA controller in closed-loop with a simulated converter in MATLAB.
6. **Hardware Integration:** Connect physical sensors (ADC interface) and complete the feedback loop.

3.1.9 Hardware-in-the-Loop (HIL) Validation

HIL testing validates controller performance in real time by executing digital logic on FPGA while the converter dynamics are simulated on MATLAB/Simulink. The exchange of input/output variables through JTAG or UART ensures closed-loop operation. This allows:

- Safe evaluation under realistic load and transient conditions,
- Verification of DPWM timing and quantization,
- Iterative fine-tuning of compensator coefficients,
- Validation of startup, overshoot, and steady-state response.

The HIL approach thus ensures controller robustness before hardware deployment.

3.1.10 Simulation Results and Comparisons

Reference Voltage Waveform

This waveform is the target output for all closed-loop control experiments. It is applied identically for voltage-mode and current-mode compensator designs to assess tracking and dynamic response.

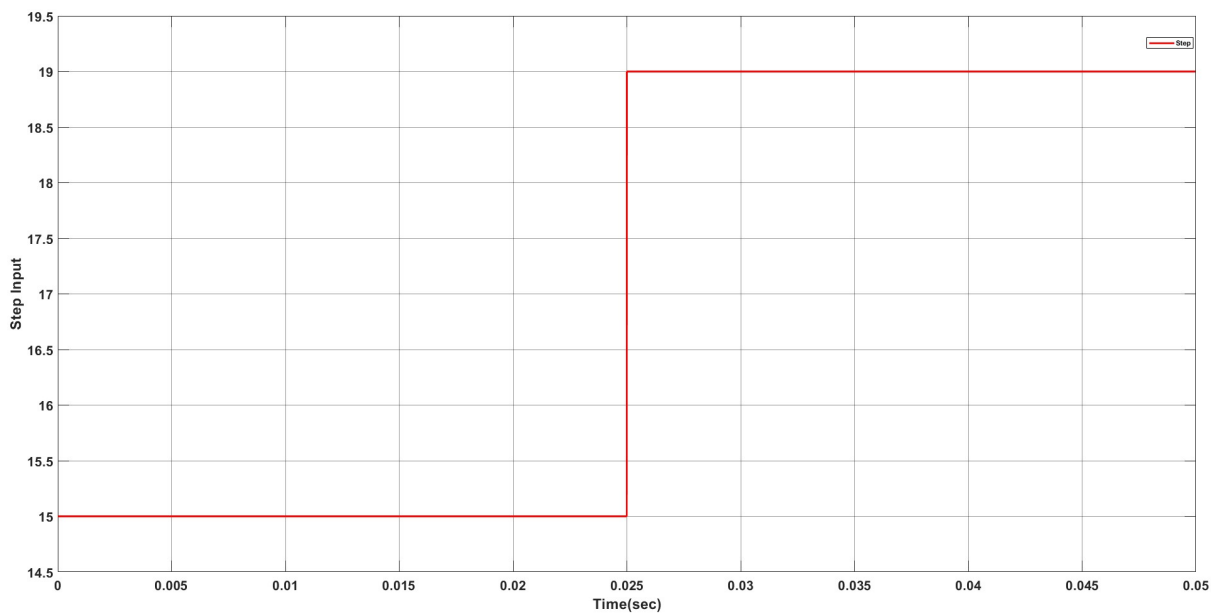


Figure 3.10: Reference voltage waveform used for all simulation and hardware-in-the-loop tests.

3.1.11 Voltage-Mode Control

S-domain vs. Z-domain Compensator Results

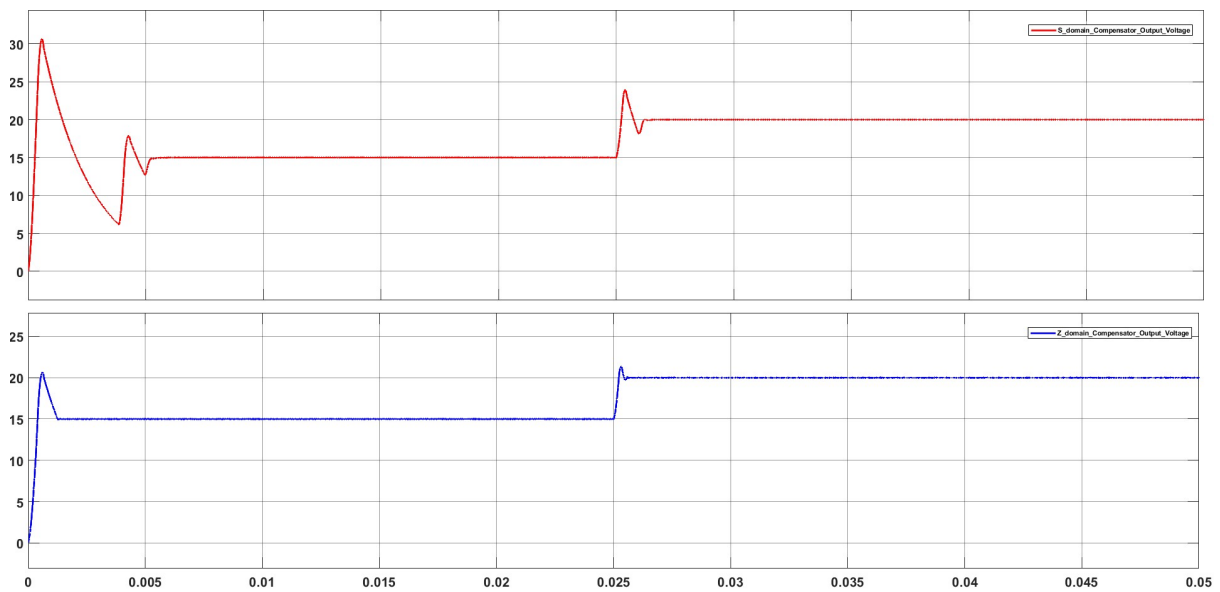


Figure 3.11: Output voltage waveform S-domain vs. Z-domain Compensator

The first graph (s-domain compensator) jumps up quickly but has more wiggles and takes some time to calm down. The second graph (z-domain compensator) rises a bit slower, but it's smoother and settles without as much movement. In the end, both get to the same steady level, but the z-domain one is more steady and gentle, which is good for digital control.

Z-domain Simulink vs. FPGA Hardware-in-the-Loop Results

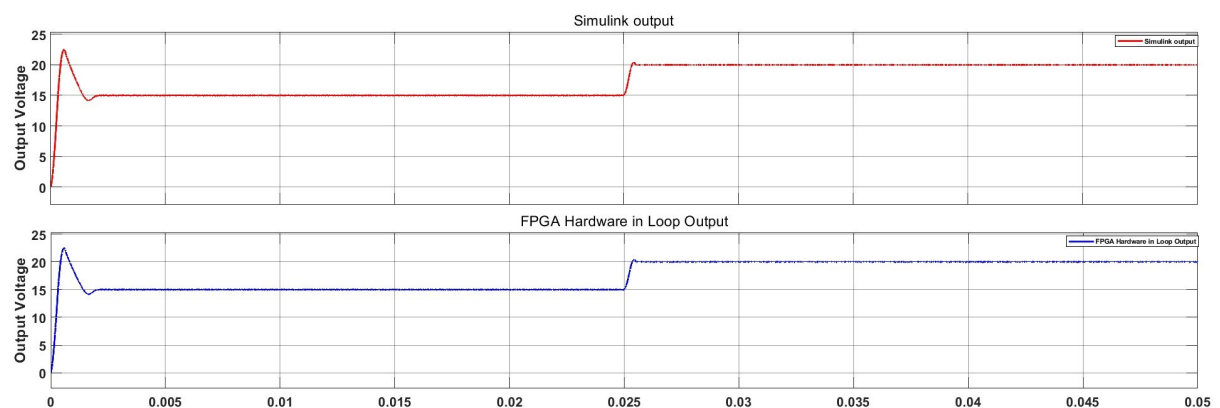


Figure 3.12: Output voltage waveform: FPGA hardware-in-the-loop implementation.

The first (top) graph shows the Simulink simulation, where the output jumps up fast and settles after a little bit of wiggling. The second (bottom) graph is the FPGA hardware in loop output. This one starts off more steady and quickly flattens out with less wobble. Both graphs end up at the same voltage in the end, but the hardware response is smoother and more stable.

3.1.12 Current-Mode Control

S-domain vs. Z-domain Compensator Results

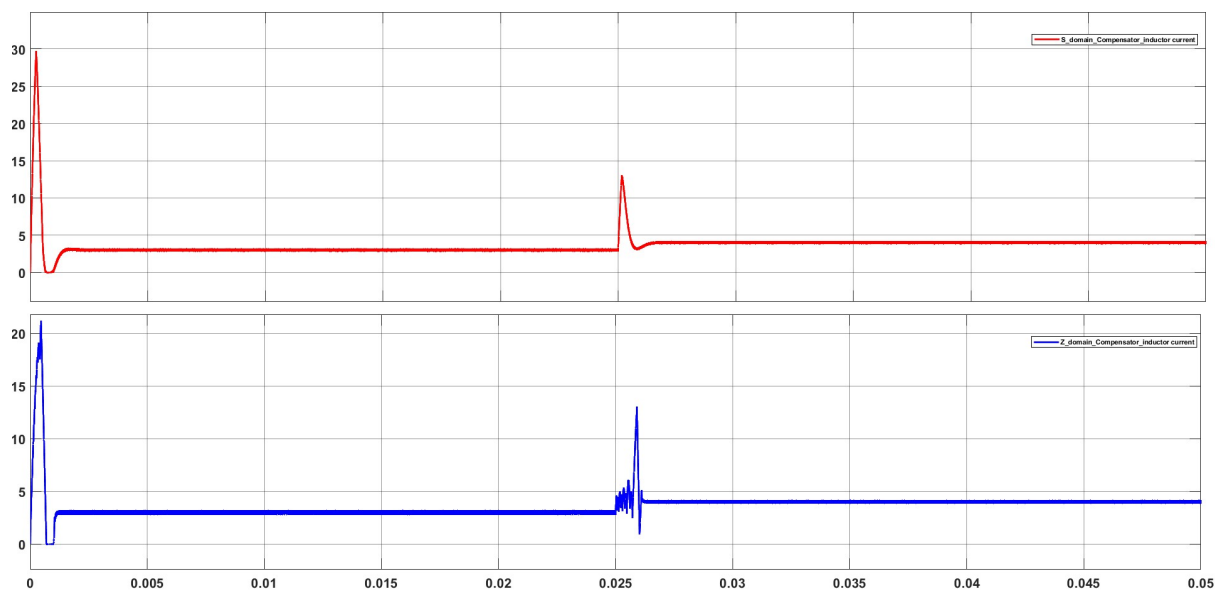


Figure 3.13: Inductor Current

Figure 3.13 presents the inductor current responses for the S-domain compensator (top) and the Z-domain compensator (bottom). Both responses demonstrate fast initial rise and settling after a step input. The S-domain output remains largely smooth following the transient, whereas the Z-domain output displays some oscillations and ripples shortly after the step before reaching steady-state.

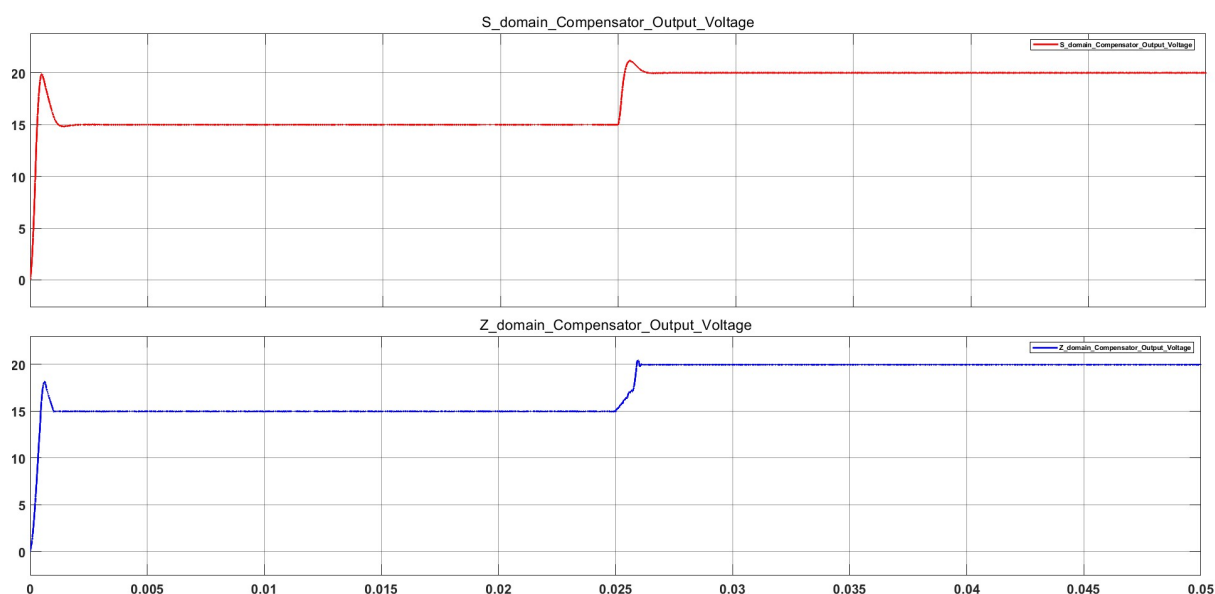


Figure 3.14: Output Voltage

The above figure compares the output voltage responses of an S-domain compensator (top)

and a Z-domain compensator (bottom). Both plots exhibit similar transient and steady-state behavior following step changes. The S-domain compensator shows a slight overshoot at the step, while the Z-domain compensator settles with minimal oscillation.

Z-domain Simulink vs. FPGA Hardware-in-the-Loop Results

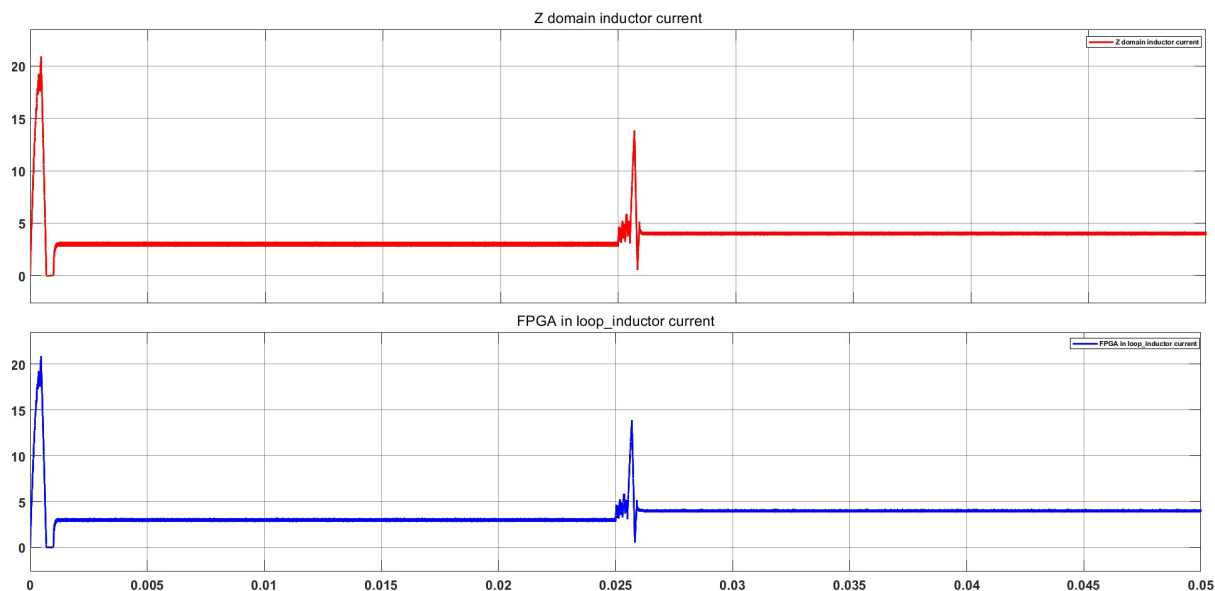


Figure 3.15: Inductor Current

These two graphs show the inductor current using a z-domain controller, with simulation results on top and FPGA hardware results below.

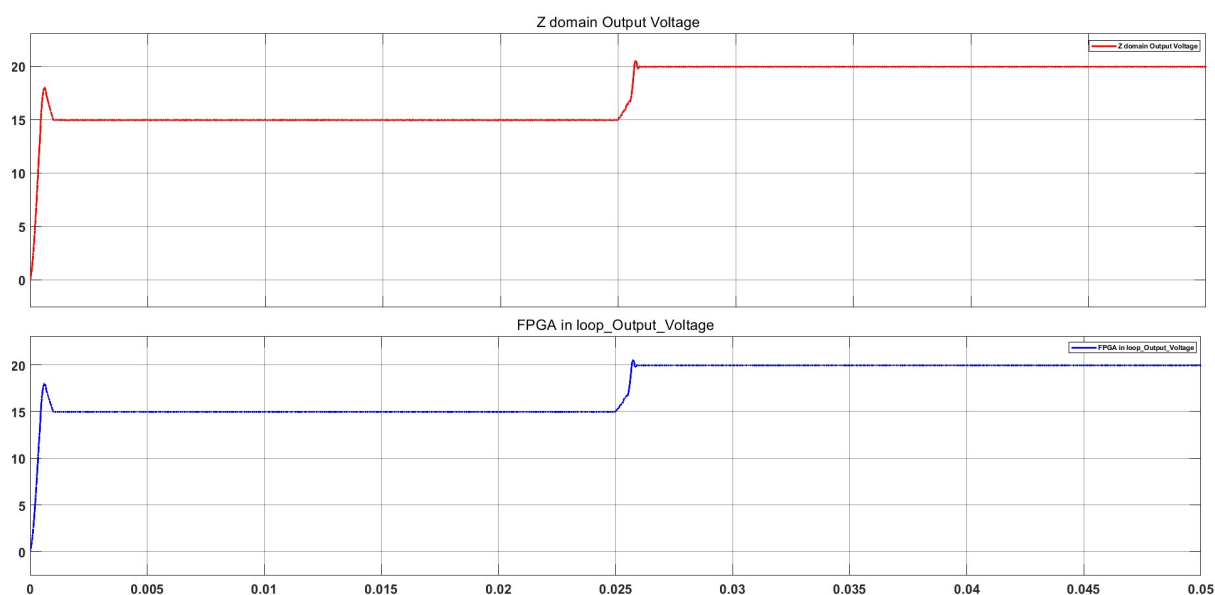


Figure 3.16: Output Voltage

These two plots show the output voltage for both the z-domain simulation (top) and FPGA hardware-in-the-loop test (bottom).

3.1.13 Summary of FPGA Control Benefits

FPGA-based prototyping provides high flexibility and precision in converter control. Features such as deterministic timing, parallel signal processing, and high-resolution PWM generation contribute to improved transient performance. Controllers developed in MATLAB can be quickly implemented and adjusted using FPGA hardware, enabling efficient testing of various control strategies. FPGA reconfigurability supports the exploration of advanced algorithms, including predictive and adaptive control, without requiring changes to physical circuits. As a result, FPGA-based digital control effectively connects theoretical design, MATLAB simulation, and practical hardware implementation for converters like Totem-Pole PFC and LLC resonant stages.

3.2 Totem-Pole PFC and Full-Bridge LLC Resonant Converter Design for EV Charging

3.2.1 System Design and Specifications

The proposed charger accepts 230 V, 50 Hz AC input and delivers a regulated 400 V DC link, subsequently it can be stepped to 800 V DC for battery charging.

Design Targets:

- Input: 230 V RMS AC, 50 Hz
- Output DC Link: 400 V DC
- Battery Output: 400 V DC
- Rated Power: 7.7 kW
- PFC Switching Frequency: 50 kHz
- Resonant Frequency (LLC): 50 kHz

3.2.2 Totem-Pole PFC Design

Overview

The Totem-Pole Power Factor Correction (PFC) converter is employed at the front end of the On-Board Charger (OBC) to ensure that the input current drawn from the single-phase AC supply is sinusoidal and in phase with the input voltage. This topology achieves high efficiency by reducing conduction losses and enabling bridgeless operation.

Operating Principle

The converter rectifies the 230 V AC input and boosts it to a regulated DC link voltage of 400 V. The Totem-Pole configuration uses two fast-switching transistors and two diodes (or GaN FETs for high-frequency operation) to achieve high efficiency and reduced reverse recovery losses. The PFC controller adjusts the duty cycle of the main switches to maintain the input current in phase with the input voltage, thereby achieving near-unity power factor.

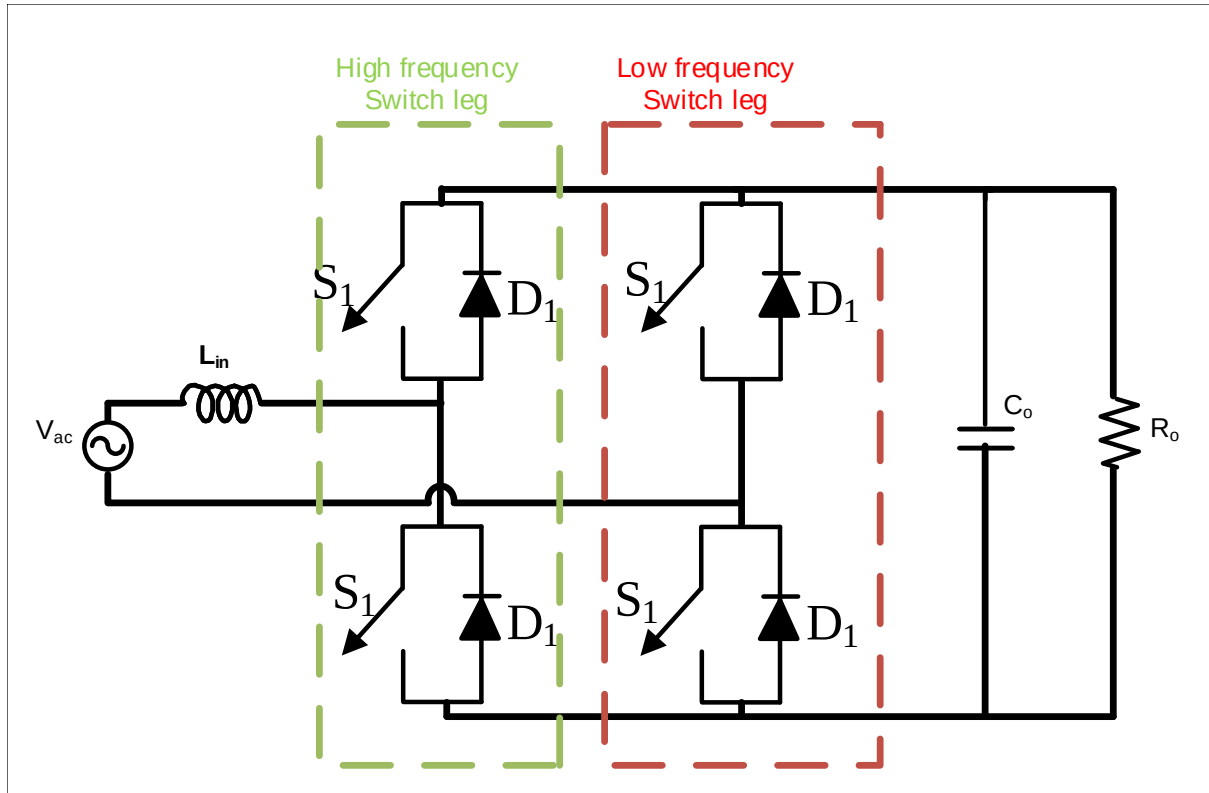


Figure 3.17: Totem-Pole PFC Circuit Diagram

Inductor Design

The boost inductor stores and transfers energy between the input source and output load. The inductor value is determined based on the desired current ripple and operating frequency:

$$L = \frac{V_{in,min} \times D}{\Delta I_L \times f_s}$$

where:

$$V_{in,min} = 230 \times \sqrt{2}/2 \text{ V}, \quad f_s = 50 \text{ kHz}, \quad D = 1 - \frac{V_{in,min}}{V_{out}}$$

and the inductor ripple current is typically chosen as:

$$\Delta I_L = (0.2 \text{ to } 0.4) \times I_{L,avg}$$

The average inductor current is:

$$I_{L,avg} = \frac{P_{out}}{V_{in,rms} \times \eta}$$

where η is the efficiency (typically 0.95–0.98).

Output Capacitor Design

The DC link capacitor filters the double-frequency ripple component of the rectified AC voltage. Its value is given by:

$$C_{out} = \frac{I_{out}}{2\omega V_{ripple}}$$

where $\omega = 2\pi f_{\text{line}}$ and V_{ripple} is the allowed output voltage ripple, typically 2–4% of the DC link voltage.

Performance Summary

For a 1-phase 230 V AC input and 400 V DC output:

$$V_{\text{in,rms}} = 230 \text{ V}, \quad V_{\text{out}} = 400 \text{ V}, \quad P_{\text{out}} = 7.7 \text{ kW}, \quad f_s = 50 \text{ kHz}$$

The designed PFC ensures a unity power factor and THD < 5%, with a regulated 400 V DC link provided as the input to the subsequent LLC stage.

3.2.3 Full-Bridge LLC Resonant Converter Design

Circuit Diagram

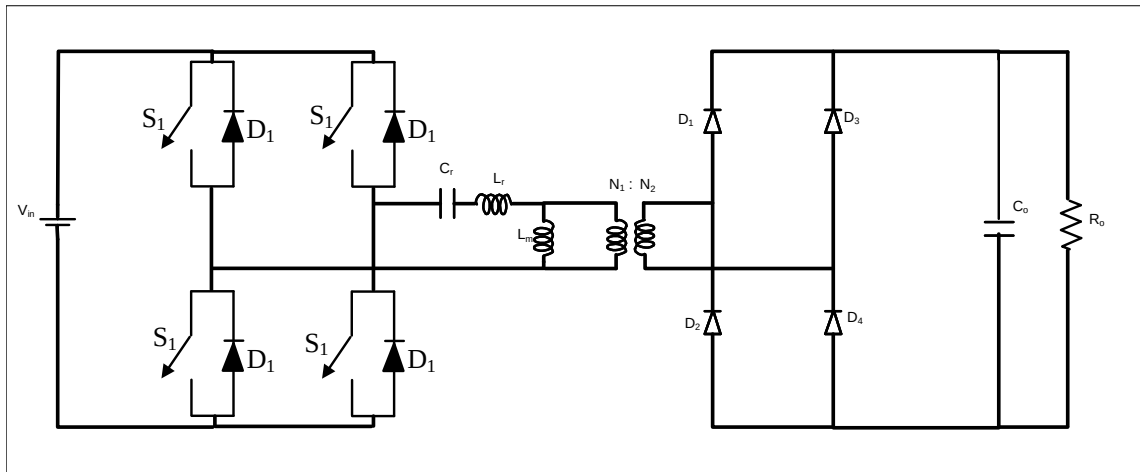


Figure 3.18: Full-Bridge LLC Resonant Converter Architecture

LLC Component Design

Resonant tank circuit of LLC converter is shown below

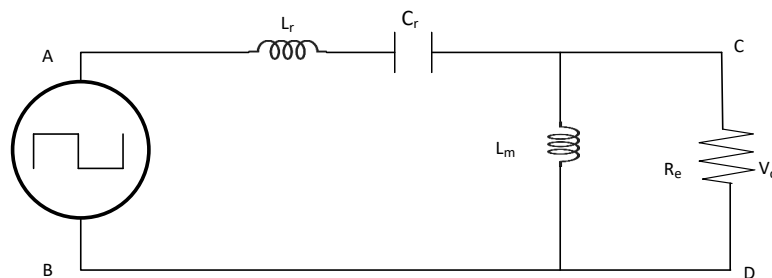


Figure 3.19: Resonant Circuit

The resonant inductor L_r , resonant capacitor C_r , and the parallel inductor the transformer's magnetizing inductor (L_m) are the three most important parts of an LLC converter for controlling the output voltage of the converter. These parts are made with the resonant frequency equal

to the switching frequency in order to function in the intended resonant mode.

The design target is only to isolate battery side.

So the transformer turns ratio is:

$$\frac{N_1}{N_2} = \frac{1}{1}$$

The small-signal voltage transfer ratio is:

$$\frac{V_{CD}}{V_{AB}} = \frac{\frac{jX_m R_e}{jX_m + R_e}}{\frac{jX_m R_e}{jX_m + R_e} + jX_{L_r} - jX_{C_r}}$$

where:

$$X_m = \omega L_m, \quad X_{L_r} = \omega L_r, \quad X_{C_r} = \frac{1}{\omega C_r}$$

and the equivalent load resistance is:

$$R_e = \frac{8}{\pi^2} \left(\frac{N_1}{N_2} \right)^2 R_L$$

The DC voltage gain is given by:

$$\frac{V_{CD}}{V_{AB}} = \frac{1}{\sqrt{1 + \frac{L_r}{L_m} \left(1 - \frac{1}{W_x^2} \right)^2 + Q^2 \left(W_x - \frac{1}{W_x} \right)^2}}$$

with:

$$Q = \frac{\omega_0 L_r}{R_e} = \frac{1}{\omega_0 C_r R_e}, \quad \omega_0 = \frac{1}{\sqrt{L_r C_r}}, \quad W_x = \frac{\omega}{\omega_0}$$

and the magnetic ratio:

$$m = \frac{L_m + L_r}{L_r}$$

The resonant circuit gain is:

$$|G(j\omega)| = \frac{W_x^2(m-1)}{\sqrt{(mW_x^2 - 1)^2 + W_x^2(W_x^2 - 1)^2(m-1)^2Q^2}}$$

A MATLAB sweep of m and Q values gives optimal parameters:

$$Q = 0.3, \quad m = 6$$

ensuring moderate gain variation and soft-switching near resonance.

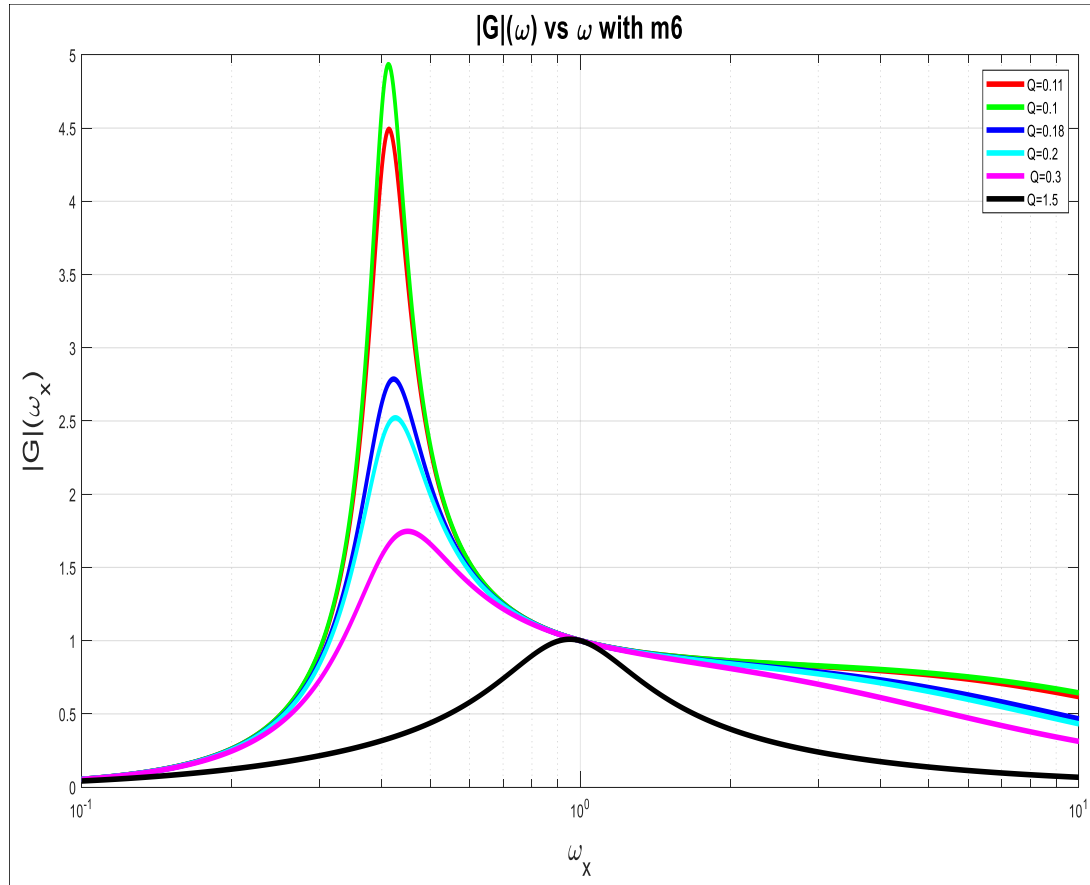


Figure 3.20: Normalized Frequency vs Gain of Resonant Tank

Resonant Element Calculation

Given:

$$V_{in} = 400 \text{ V}, \quad V_{out} = 400 \text{ V}, \quad P = 7.7 \text{ kW}, \quad f_{sw} = 50 \text{ kHz}$$

Load resistance:

$$R_L = \frac{V_{out}^2}{P} = 20.77 \Omega$$

Reflected load resistance:

$$R_e = \frac{8}{\pi^2} \left(\frac{400}{48} \right)^2 R_L = 16.835 \Omega$$

Resonant parameters (example calculations):

$$L_r = \frac{Q R_e}{\omega_0} = 16.07 \mu H$$

$$C_r = \frac{1}{\omega_0^2 L_r} = 630 \text{ nF}$$

$$L_m = (m - 1) L_r = 80.35 \mu H$$

3.2.4 Simulation Setup and Results

Table 3.2: LLC Resonant Converter Parameters Used in MATLAB Simulation

Parameter	Symbol	Value	Unit
Input Voltage	V_{in}	400	V
Output Voltage	V_{out}	400	V
Output Power	P	7.7	kW
Switching Frequency	f_{sw}	50	kHz
Load Resistance	R_L	20.77	Ω
Resonant Inductance	L_r	16.07	μH
Resonant Capacitance	C_r	630	nF
Magnetizing Inductance	L_m	80.35	μH

Simulation Circuit

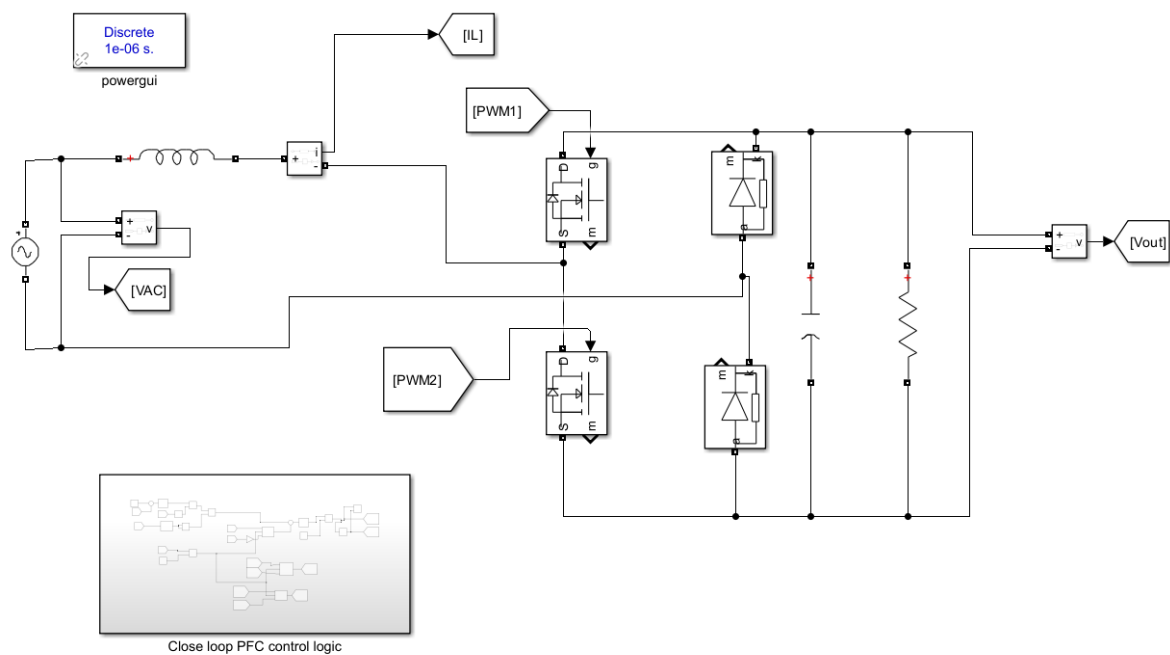


Figure 3.21: Totem-Pole PFC Circuit

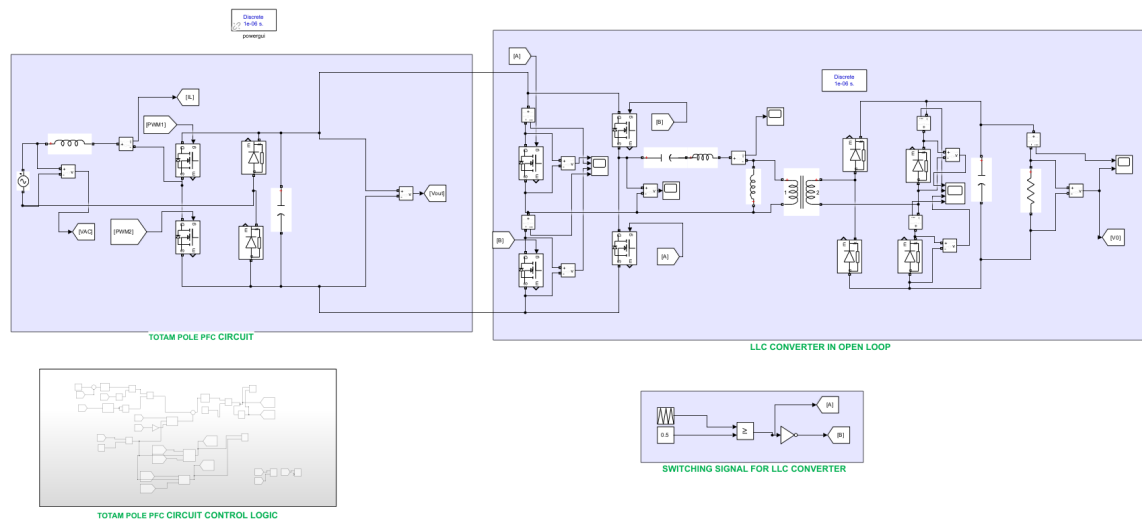


Figure 3.22: Circuit diagram of proposed converter

Waveforms

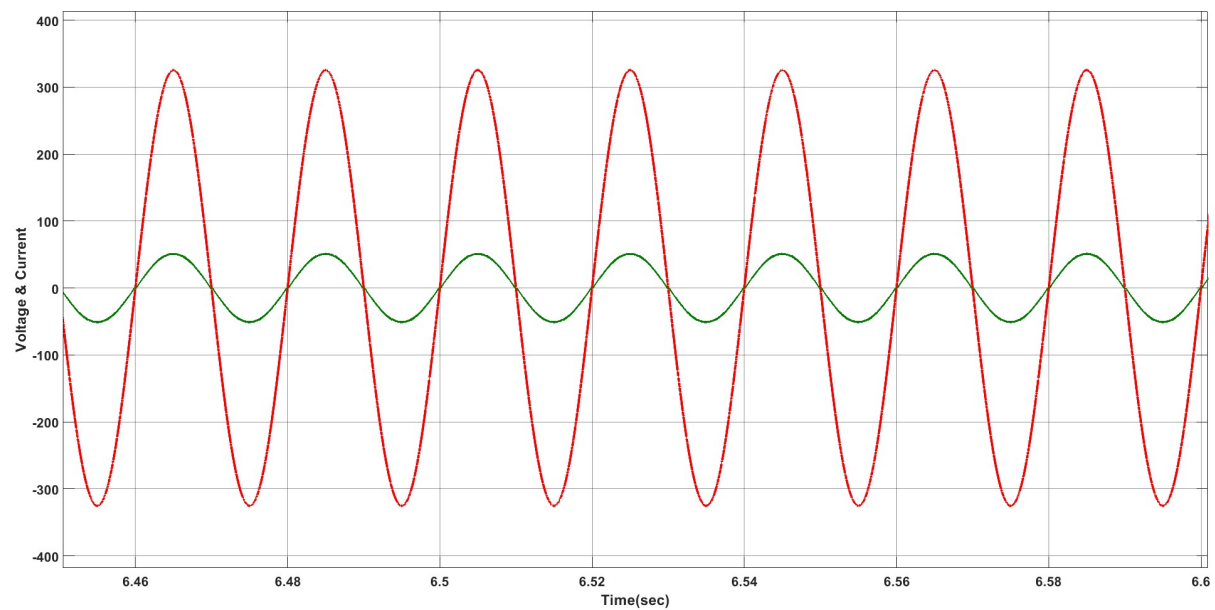


Figure 3.23: AC Side Input Voltage And Current

Above given plot reflects input voltage as in phase with input current.

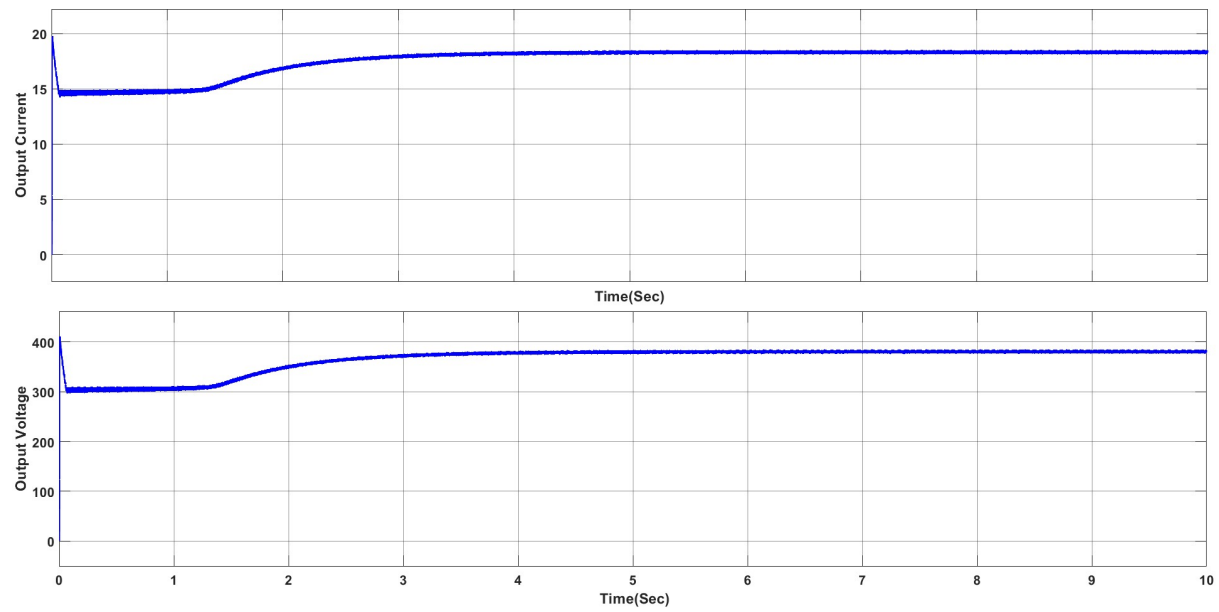


Figure 3.24: Output Voltage and Current

The output plot is observed and it reflects the output voltage of 385V and current of 18.2A. Since the simulation is open loop there is a marginal steady state error in the output waveform.

4 Conclusion and Future Work

4.1 Conclusion

The proposed charger uses a **Totem-Pole Power Factor Correction (PFC)** stage with a **Full-Bridge LLC Resonant DC–DC Converter**, forming an efficient, compact system for both 400 V and 800 V battery electric vehicles [3]. Simulation and analytical checks confirm that the Totem-Pole PFC keeps the power factor near unity, reduces grid pollution ($\text{THD} < 5\%$), and the LLC converter provides reliable isolation and achieves soft-switching for better efficiency [4].

A key part of this work is the **FPGA-based digital control implementation**. Using an FPGA enables much faster and more flexible regulation of both PFC and LLC stages compared to traditional microcontrollers. Benefits include low control latency, precise timing, parallel real-time computation, and on-the-fly reconfiguration of control laws for changing grid or load conditions. Hardware-in-the-loop experiments demonstrated closed-loop stability with tight voltage and current regulation.

Comparing the LLC resonant converter with alternatives like the Dual Active Bridge (DAB):

- LLC offers **higher efficiency at typical EV charger loads** because it uses soft-switching (ZVS/ZCS), which minimizes switching and conduction losses.
- The magnetic components in LLC are smaller, letting the charger be more compact and lighter.
- LLC is **easier to design** for wide input voltage variation (like 400 V and 800 V batteries) and naturally suppresses EMI, unlike DAB which can generate more harmonics unless carefully tuned.
- DAB topologies are best for very high-power, bidirectional systems (V2G), but for most onboard chargers, LLC delivers better cost and performance.

By combining the strengths of advanced power converter topologies, wide-bandgap devices, and fast FPGA-based control, this charger is well suited for next-generation EV platforms—delivering high efficiency, compact size, reliable operation, and full compatibility with today’s international standards [8].

4.2 Future Work and Control Implementation

There are many ways to improve this system and make it even more advanced:

1. **Digital Control:** Add smart controllers using DSP or FPGA chips, so the charger can easily adapt to changes in the power grid or battery.
2. **Better Algorithms:** Try new control methods like model predictive control or sliding mode control to help the charger respond faster and stay stable.
3. **High-Speed Switching:** Explore “soft-switching” techniques that let the charger work at even higher frequencies (over 300 kHz) and boost efficiency.
4. **Wide-Bandgap Devices:** Test real prototypes with GaN or SiC components to see the benefits like cooler operation and lower EMI.

5. **Hardware Tests:** Build and measure a physical charger to prove its efficiency and safe operation.
6. **Bidirectional Charging:** Expand the charger to support sending power back to the grid or a house (V2G/V2H).
7. **Reliability Studies:** Run thermal and reliability checks to make sure the charger stands up to tough automotive standards.

With these improvements, the charger can become truly next-generation—smarter, faster, and ready for the future of electric vehicles.

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